

10.9 SYMMETRY FACTORS

In quantum field theory, symmetry factors correct for overcounting in perturbative expansions. This note supplements the discussion in lectures: first the definition and three standard examples, then worked examples in φ^3 theory.

ORIGIN AND DEFINITION

Symmetry factors arise because several *Wick contractions*, or permutations of vertices and lines, can produce the same term. The redundancy is measured by the **automorphism group** of the diagram: all relabelings of vertices and of the line ends attached to them (permutations, rotations, reflections) that leave the diagram unchanged. The **symmetry factor** is the order of this group, $S = |\text{Aut}(G)|$, and each diagram is weighted by $1/S$.

THREE EXAMPLES

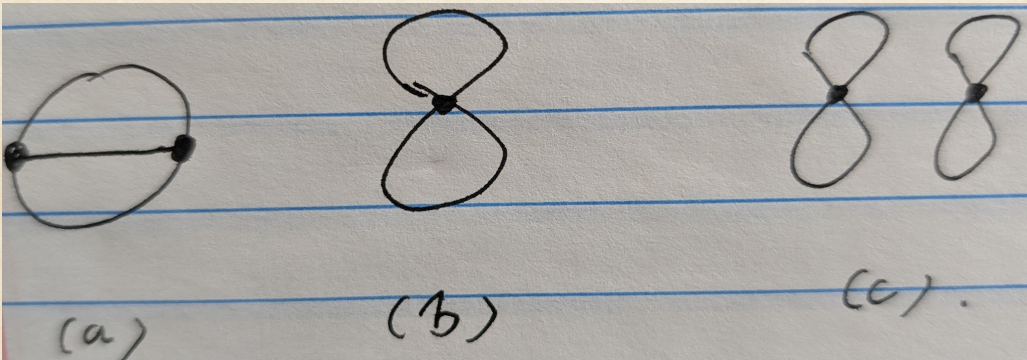


Diagram (a) is in φ^3 theory; (b) and (c) are vacuum diagrams of φ^4 theory, where each vertex has four lines.

- **The “sunset” diagram in φ^3 theory (a):** two vertices connected by three lines.
 - Symmetries: vertex swap ($\mathbb{Z}_2$) and permutations of the three lines (S_3).
 - $|\text{Aut}(G)| = 2 \times 6 = 12 \Rightarrow S = 12$.
- **The vacuum “figure-eight” in φ^4 theory (b):** one vertex with two loops.
 - Symmetries: swapping the loops ($\mathbb{Z}_2$) and flipping each loop ($\mathbb{Z}_2 \times \mathbb{Z}_2$).
 - $|\text{Aut}(G)| = 2 \times 2 \times 2 = 8 \Rightarrow S = 8$.
- **Two identical disconnected pieces (c):** two copies of the figure-eight.

- 1552 – Symmetries: swapping the two pieces ($\mathbb{Z}_2$) and the
 1553 symmetries of each piece.
 1554 – $|\text{Aut}(G)| = 2 \times (8 \times 8) = 128 \Rightarrow S = 128$.

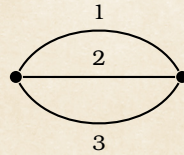
1555 WORKED EXAMPLES IN φ^3 THEORY

Recall how we “got rid of” the factor

$$\frac{1}{V!} \left(\frac{1}{3!} \right)^V \frac{1}{P!} \left(\frac{1}{2!} \right)^P$$

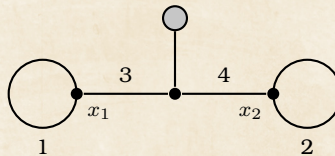
1556 in the expansion of $Z(J)$ (Srednicki §9). We treated the V vertices and the
 1557 P propagators as identical and exchangeable, which gives $V! P!$
 1558 rearrangements. The three derivatives $\left(\frac{1}{i} \frac{\delta}{\delta J} \right)^3$ of each vertex, acting on the
 1559 factors $\frac{1}{2} J \Delta J$, then supply another $(3!)^V (2!)^P$ possibilities, cancelling the
 1560 factor exactly. But this overcounts whenever some of those
 1561 rearrangements give back the same term. The overcounting is the
 1562 symmetry factor S , and each diagram is weighted by $1/S$. Shaded circles
 1563 below are sources.

1564 EXAMPLE A: THE SUNSET, BY COUNTING



- 1565
- 1566 (1) The $(3!)^2$ permutations of the two $\left[\frac{1}{i} \frac{\delta}{\delta J} \right]^3$ give only $3!$ different
 1567 products Δ^3 if one distinguishes the propagators. That is an
 1568 overcounting by $3!$.
- 1569 (2) The $(2!)^3$ flips of the propagators are already equivalent to
 1570 exchanging the two vertices, which was counted in $V! = 2!$. That is
 1571 an overcounting by 2.
- 1572 (3) The total overcounting is $2 \times 3! = 12$, so $S = 12$.
- 1573 (4) The same number follows from the symmetries of the diagram, as
 1574 for diagram (a) above: the three propagators can be permuted ($3!$),
 1575 and the two vertices can be exchanged by a reflection (2).

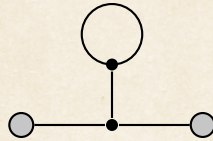
1576 EXAMPLE B: TWO TADPOLES ON A LINE, ONE SOURCE



- 1577
- 1578 (1) The loops 1 and 2 are $\Delta(x_1 - x_1)$ and $\Delta(x_2 - x_2)$. Flipping the ends of
 1579 a loop gives back the same term; there is no $\Delta(x - y) \rightarrow \Delta(y - x)$ to
 1580 tell the two orders apart. So the weight is overcounted by 2×2 .

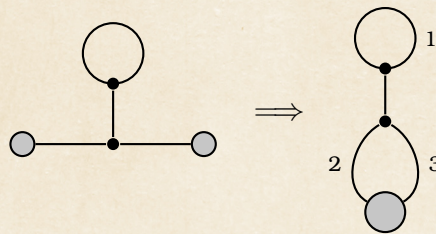
- 1581 (2) Exchanging propagators 3 and 4 together with the vertices x_1 and
 1582 x_2 gives the same process. It only changes how the functional
 1583 derivatives are arranged and produces no new term, so it
 1584 overcounts by another 2.
 1585 (3) The symmetry factor is $S = 2 \times 2 \times 2 = 8$.
 1586 (4) By symmetry: a left-right reflection, and an up-down reflection of
 1587 each of the two loops, give $S = 2 \times 2 \times 2 = 8$ again.

1588 **EXAMPLE C: A TADPOLE ON A LINE BETWEEN TWO SOURCES**



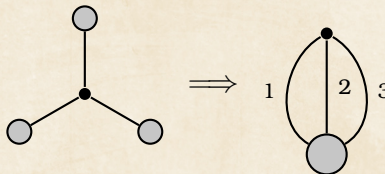
- 1590 (1) The loop gives an overcounting of 2.
 1591 (2) The left-right reflection gives another 2.
 1592 (3) $S = 2 \times 2 = 4$.

1593 For several sources there is a useful *deformation method*. Since the
 1594 sources are identical, deform the diagram so that all of them meet in one
 1595 shaded blob. The blob is not a vertex: any number of lines may end on it.
 1596 Then read off the symmetries of the deformed diagram. For Example C:



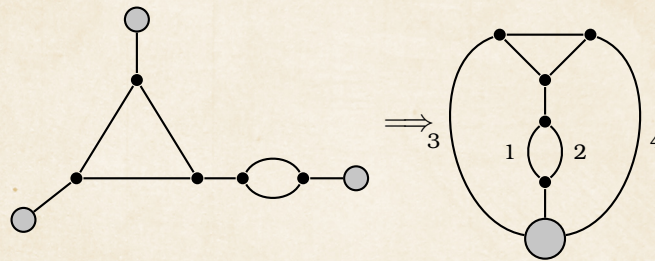
1598 Propagator 1 gives a factor of 2, and propagators 2 and 3 can be
 1599 exchanged (2!). So $S = 2 \times 2 = 4$.

1600 **EXAMPLE D: THREE SOURCES ON ONE VERTEX**



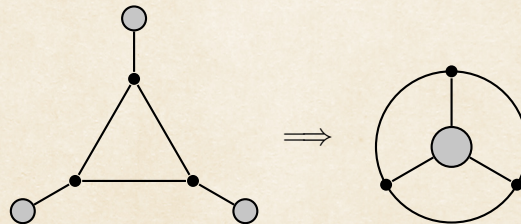
1602 The three propagators can be permuted in every way, so $S = 3! = 6$.

1603 **EXAMPLE E: A TRIANGLE WITH A BUBBLE ON ONE LEG**



1604
1605 After the deformation one can exchange $1 \leftrightarrow 2$ (the two lines of the bubble)
1606 and $3 \leftrightarrow 4$ (a reflection of the triangle that fixes the dressed corner). So
1607 $S = 2 \times 2 = 4$.

1608 **EXAMPLE F: A TRIANGLE WITH THREE BARE LEGS**



1609
1610 The deformed diagram has a cyclic symmetry of order 3 (rotations by 120°
1611 and 240°) and a reflection. The three reflections, one through each corner,
1612 are not independent: each is a rotation times one reflection. So
1613 $S = 3 \times 2 = 6$.

1614 **SUMMARY**

The contribution of a diagram to the amplitude is

$$\text{Contribution} = \frac{1}{S} \times (\text{couplings}) \times (\text{integrals}).$$

1615 Symmetry factors ensure that each distinct term is counted exactly once.
1616 Errors in S propagate to observables such as cross sections, so getting
1617 them right matters. I directly write down the symmetry factors of many
1618 diagrams in my lecture notes and provide additional notes explaining
1619 different ways to get them. There is ultimately one method: carry out the
1620 functional-derivative expansion of the path integral. There, you won't
1621 need to introduce symmetry factors, but rather count how many ways
1622 there are to get the identical expression.